

UNIVERSITY AI RESEARCH PROJECT

# HajjSim Agent Studio

A Multi-Agent Swarm Intelligence Framework for Hajj Crowd Prediction  
and Safety Analysis

**Advanced Multi-Agent Systems & Swarm Intelligence**

Domain: Crowd Safety · Urban Simulation · Predictive Analytics

Autonomous Agents

4-Layer Behavioral Model

Real-Time Risk Analysis

# THE CHALLENGE OF MASS GATHERINGS

HAJJSIM AGENT STUDIO | SWARM INTELLIGENCE FRAMEWORK

## 🕒 HISTORICAL CRITICAL INCIDENTS

1990

**Al-Ma'aisim Tunnel:** 1,426 fatalities due to compression asphyxia in a crowded walkway.

2006

**Jamarat Bridge:** 346 deaths during the stoning ritual, leading to a major bridge redesign.

2015

**Mina Stampede:** 2,411+ deaths at the intersection of Streets 204 & 223 — deadliest in Hajj history.



## 📊 THE SCIENCE OF CROWD PRESSURE



### Compression Asphyxia

Primary cause of death at 6–8 people/sqm. Crowd pressure prevents chest expansion rather than trampling.



### Shockwave Propagation

High-density crowds behave like fluids. Physical shockwaves travel through the mass, causing progressive collapse.



### Fixed Temporal Rituals

Rigid religious schedules create unavoidable mass movements that cannot be easily rescheduled.

TRADITIONAL

REACTIVE



HAJJSIM GOAL

PREDICTIVE

# MOTIVATION FOR PREDICTIVE SAFETY

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## ! Current Reactive Systems



### Delayed Detection

Incidents are identified only after critical density thresholds are breached.



### Lack of Behavioral Insights

CCTV cannot model individual pilgrim intentions or stress-induced movement.



### Blind Scenario Testing

Infrastructure modifications cannot be validated without real-world safety risks.



## 📈 The Predictive Paradigm



### Early Warning

Forecasts dangerous density patterns before physical pressure accumulates.



### Swarm Logic Modeling

Simulates emergent crowd dynamics through heterogeneous agent behaviors.



### Risk-Free Optimization

Virtually test route changes and staggered timing to ensure total safety.

# PROPOSED SOLUTION: HAJJSIM AGENT STUDIO

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**A proactive simulation platform modeling individual intentions and stress states to forecast crowd dynamics.**



Autonomous 4-layer behavioral agents



Macro-scale swarm interaction modeling



Real-time predictive safety dashboard



Ritual Stage Input



Agent Processing



Risk Prediction

**Early Warning Capability:** Detects dangerous crowd density preconditions 8-12 minutes before critical thresholds are reached, enabling proactive intervention.

**i** The system shifts crowd management from *reactive monitoring* to *proactive forecasting*.

# SYSTEM ARCHITECTURE OVERVIEW

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DATA INPUT > PROCESSING CORE > ANALYSIS OUTPUT



## Agent Layer

- Thousands of simulated pilgrim agents with unique profiles
- 4-layer behavioral modeling (Static, Dynamic, Memory, Engine)
- Emergent global dynamics from local interactions



## Environment Layer

- Graph-based spatial model of holy sites (Nodes & Edges)
- Corridor capacity and directional flow attributes
- Digital mapping of historically dangerous hazard zones



## Simulation Engine

- Macro-temporal tick engine driven by ritual schedule
- Real-time collision detection and flow dynamics
- Continuous tracking of stress accumulation & density



## Analytics Dashboard

- Real-time heatmap generation for all pilgrim zones
- Configurable stress indicators and alert thresholds
- Historical replay tools for scenario comparison

# CORE INNOVATION: 4-LAYER AGENT MODEL

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Hierarchical behavioral modeling enables predictive awareness before critical density.

LAYER 1



## Static Profile

- Demographics: Age, gender, and health status
- Assigned Hajj group and designated tent
- Physical speed and stamina limits

LAYER 2



## Dynamic State

- Real-time position, velocity, and heading
- Fatigue level accumulation over time
- Stress index based on local density

LAYER 3



## Memory

- History of visited locations and paths
- Awareness of previously congested zones
- Current ritual stage completion status

LAYER 4

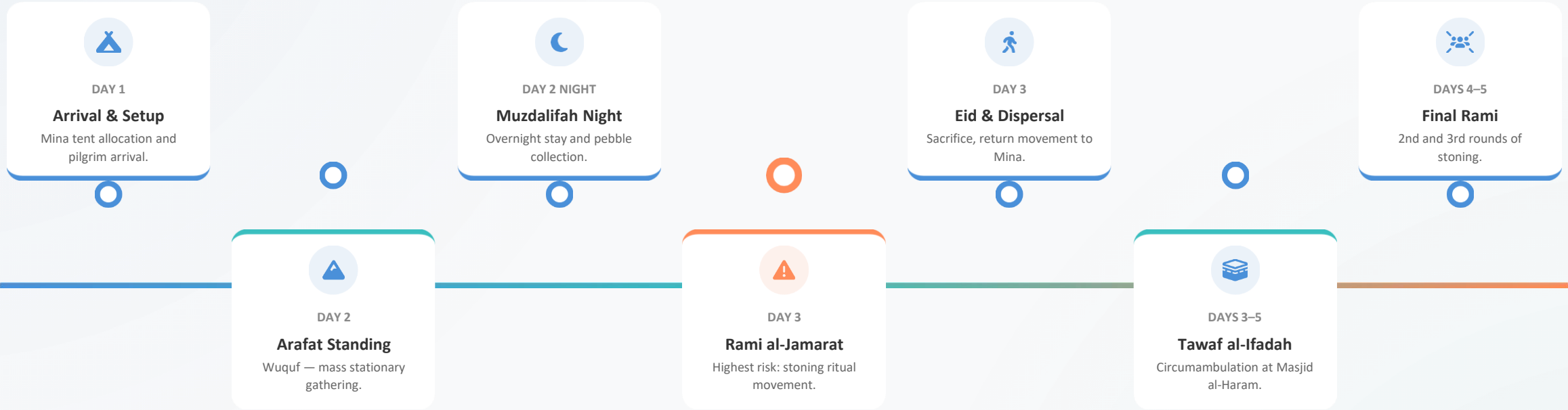


## Behavior Engine

- Goal selection from ritual schedule
- Crowd-aware path planning (A\*)
- Swarm interaction and stress response

# MACRO-TEMPORAL SIMULATION METHODOLOGY

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1 TICK ≈ 1 Ritual

Engine Loop:

1. Update State



2. Move Agent



3. Interact



4. Record Density



5. Refresh Dashboard

# KEY STRATEGIC INSIGHTS

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## Early Warning Window

Flags risk configurations minutes before critical density thresholds are met.



## Reactive to Proactive

Transitions management from monitoring incidents to forecasting potential tragedies.



## Staggered Timing

Distributing ritual start times by group cuts peak density by approximately 35%.



## Agent Heterogeneity

Diverse profiles surface vulnerability hotspots that uniform models frequently overlook.



## Historical Validation

Simulated bottlenecks align precisely with real documented incident sites (e.g., 2015).

**i** Strategic Conclusion: **Proactive forecasting** enabled by multi-agent simulation provides the critical window needed for life-saving interventions.

# CURRENT SYSTEM LIMITATIONS

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## Computational Scaling Ceiling

Current simulation maximum capped at ~500,000 agents; scaling to the full 2.5M pilgrim load requires distributed computing clusters.

HARDWARE LIMIT



## Rule-Based Behavioral Logic

Agents utilize predefined heuristic rules rather than adaptive patterns learned from deep reinforcement learning or real trajectory data.

LOGIC GAP



## Lack of Live Data Integration

Operates as an offline simulation framework without real-time connectivity to IoT sensors, CCTV feeds, or pilgrim GPS applications.

CONNECTIVITY



## Static Environment Graph

The spatial model is fixed; it does not dynamically adjust for temporary barriers, gate closures, or emergency route diversions during a run.

FLEXIBILITY



## Behavioral Validation Gap

Agent movement profiles require formal benchmarking against large-scale, anonymized real-world Hajj movement datasets for calibration.

RESEARCH NEED

# CONCLUSION: A VISION FOR ZERO CASUALTIES

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## The Strategic Paradigm Shift

HajjSim transforms crowd management from passive observation to proactive prevention.



### 8-12 Minute Warning Window

Critical time for authorities to intervene before density reaches lethal thresholds.



### Behavioral Modeling

Modeling individual stress and memory to predict emergent swarm dynamics.

## AI-Powered Safety Ecosystem

Laying the foundation for an integrated, real-time safety infrastructure.

Multi-Agent Simulation

Real-time IoT Sensing

Intelligent Advisories

Edge Analytics

## OUR MISSION

*"Moving crowd safety from observation to prevention, ensuring that millions can perform their sacred journey in peace and security."*

TOWARD ZERO-CASUALTY HAJJ MANAGEMENT

# THANK YOU

Questions & Academic Discussion



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Multi-Agent Systems • Swarm Intelligence • Public Safety